

MAURYA PANKAJ

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SUMMARY

B.Tech graduate in Artificial Intelligence and Machine Learning with hands-on experience in machine learning, computer vision, deep learning, NLP, and Retrieval-Augmented Generation (RAG). Skilled in Python, Scikit-learn, PyTorch, TensorFlow, YOLOv8, OpenCV, LangChain, FAISS, Pandas, NumPy, MLflow, Docker, and Linux. Experienced in data preprocessing, feature engineering, model training, evaluation, optimization, experimentation, and deployment. Seeking entry-level AI/ML Engineer, Machine Learning Engineer, or Computer Vision Engineer roles.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning 2022 – 2026
NIMS University, Rajasthan, Jaipur, India

TECHNICAL SKILLS

Programming	Python, C
Machine Learning	Scikit-learn, Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Hyperparameter Optimization, GridSearchCV
Deep Learning	PyTorch, TensorFlow, Keras, Neural Networks
Computer Vision	YOLOv8, OpenCV, Object Detection, Image Classification, ONNX
NLP and LLM	NLP, RAG, LangChain, Hugging Face, LLMs, FAISS, Embeddings, Semantic Search
Data Science	Pandas, NumPy, Matplotlib
MLOps and Deployment	MLflow, Docker, FastAPI, Streamlit, ONNX
Cloud and Systems	AWS EC2, AWS S3, Linux, Git, Google Colab

EXPERIENCE

TECHFELLOWS – Machine Learning Intern Oct 2025 – Apr 2026
Remote

- Developed end-to-end machine learning workflows for data preprocessing, feature engineering, model training, and evaluation using Python, Scikit-learn, and Pandas on structured datasets containing 50,000+ records.
- Tracked 10+ machine learning experiments using MLflow, enabling reproducible training and comparison of model performance across hyperparameter configurations.
- Reduced data preprocessing runtime by approximately 30% through vectorized NumPy operations and optimized Pandas workflows.
- Contributed to AI/ML research and development through model validation, experimentation, and deployment documentation.

PROJECTS

AI-Based Research Paper Review Assistant 2025
Python, LangChain, RAG, LLM, FAISS, Streamlit

- Developed a Retrieval-Augmented Generation (RAG) application for semantic search and question answering across 100+ research papers.
- Built a PDF ingestion pipeline for document extraction, text chunking, embedding generation, and vector retrieval using FAISS.
- Integrated LLM-based response generation to provide context-aware answers from retrieved research content.
- Developed a Streamlit interface for document upload, summarization, and question answering, reducing manual review time by approximately 60%.

Real-Time Forest Fire Detection System

2024

Python, YOLOv8, PyTorch, OpenCV, NVIDIA GPU, ONNX

- Trained a YOLOv8 fire and smoke detection model on a custom dataset containing 2,000+ images, achieving 91% mAP@0.5 on validation data.
- Deployed the trained model on an NVIDIA GPU for real-time video inference at 28 FPS during testing.
- Implemented confidence-threshold filtering to reduce false positives in low-light and foggy environments.
- Built an end-to-end computer vision pipeline covering data labeling, model training, evaluation, and ONNX export for cross-platform inference.

Breast Cancer Detection Using Machine Learning

2024

Python, Scikit-learn, Pandas, NumPy, Matplotlib

- Built and evaluated SVM, Random Forest, Logistic Regression, and KNN classifiers using the Wisconsin Breast Cancer dataset containing 569 samples.
- Achieved 98.2% classification accuracy using Random Forest after hyperparameter optimization with GridSearchCV.
- Performed exploratory data analysis and feature selection using correlation analysis, reducing the feature space by 40% without reducing model accuracy.

CERTIFICATIONS

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| • Oracle AI Foundation Associate Certification – Oracle | 2025 |
| • HackerRank Python (5-Star) – HackerRank | 2025 |
| • Machine Learning Internship Certificate – TECHFELLOWS | 2026 |